# SCX Mathematical Theory: Unified Theorem Document

**Version**: 2026-06-28 | **Status**: Complete | **Audit**: Post-verification **Purpose**: Single source of truth for all 6 theorems in the SCX theory chain, for Nature SI submission. **Verification status**: All formulas checked for sign errors, all inequality directions verified, all constants traced to definitions, all cross-file notation conflicts resolved.

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## Table of Contents

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# 0. Preface

## 0.1 Unified Notation Glossary

All symbols used across the 6 theorems are collected here. Conflicting uses are disambiguated with subscripts.

| Symbol | Meaning | Used in | Notes |
|---|---|---|---|
| $\mathcal{X}$ | Input space | Thm 1,2,3,5 | |
| $\mathcal{Y}$ | Label space | Thm 1,3 | $\lvert\mathcal{Y}\rvert = K_{\mathcal{Y}}$ |
| $K_{\mathcal{Y}}$ | Number of classes | Thm 1,3 | Disambiguated from $K_{\mathcal{S}}$ |
| $\mathcal{S}$ | State space (index set) | Thm 1,2,3 | $\lvert\mathcal{S}\rvert = K_{\mathcal{S}}$ |
| $K_{\mathcal{S}}$ | Number of states | Thm 2,5 | Disambiguated from $K_{\mathcal{Y}}$ |
| $K$ | Generic state/class count | *various* | **Always disambiguated in this document** |
| $s(x) : \mathcal{X} \to \mathcal{S}$ | True state assignment | Thm 1,2,3,5 | Unobserved during clustering |
| $\hat{s}(x)$ | Estimated state assignment | Thm 2,5 | From k-means on $\phi(x)$ |

| Symbol | Meaning | Used in | Notes |
|---|---|---|---|
| $\Pi = \{s_1, \dots, s_{K_s}\}$ | State partition of $\mathcal{X}$ | Thm 1 | Measurable partition |
| $\rho_s = \mathbb{P}(X \in s)$ | State probability | Thm 1 | $\sum_s \rho_s = 1$ |
| $\mu_s$ | State-$s$ clean error upper bound | Thm 1 | $\sup_{x \in s} \mathbb{E}[C \mid \text{clean}, X = x] \leq \mu_s$ |
| $C_{\text{bal}}$ | Error balance constant ($\geq 1$) | Thm 1 | Controls error concentration |
| $f^* : \mathcal{X} \to \mathcal{Y}$ | True oracle | Thm 1,3 | Unobserved |
| $\{f_m\}_{m=1}^M$ | Expert models | Thm 1,2,3 | $M$ experts |
| $e_m(x, y) = \mathbf{1}\{\ell(f_m(x), y) > \tau\}$ | Expert error indicator | Thm 1 | |
| $C(x) = \frac{1}{M} \sum_m e_m(x, y)$ | Consensus score | Thm 1 | |
| $\theta$ | Detection threshold | Thm 1,4' | |
| $\eta$ | Global noise rate | Thm 1,2,3,4' | $\eta \in (0, 1/2)$ typically |
| $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, B]$ | Bounded loss | Thm 1 | |
| $\tau > 0$ | Expert error threshold | Thm 1 | |
| $\Delta_s$ | State-$s$ separation gap | Thm 1 | |
| $\phi(X)$ | Feature representation | Thm 2,5 | $\phi : \mathcal{X} \to \mathbb{R}^{d_\phi}$ |

| Symbol | Meaning | Used in | Notes |
|---|---|---|---|
| $\delta = I(\phi(X); S)$ | Feature-state mutual info | Thm 2 | In nats |
| $\varepsilon_\phi = \delta / \log K_{\mathcal{S}}$ | Normalized feature weakness | Thm 2 | |
| $Z \in \{0, 1\}$ | Noise indicator | Thm 2 | $Z = 1$ for noise |
| $D_m(x) = \ell(f_m(x), y)$ | Expert loss | Thm 2 | |
| $NS(x)$ | SCX noise score | Thm 2 | |
| $C_F$ | F1 Lipschitz constant | Thm 2 | $C_F \leq 2$ in operating range |
| $p_0 = \mu_s$ | Clean error rate (state $s$) | Thm 4' | |
| $p_1 = 1 - C_{\text{bal}} \cdot \mu_s/(K_y - 1)$ | Noise error rate (state $s$) | Thm 4' | |
| $\kappa = C(\text{Bern}(p_0), \text{Bern}(p_1))$ | Chernoff information | Thm 4' | $\kappa = \text{KL}(\theta^* \Vert p_0) = \text{KL}(\theta^* \Vert p_1)$ |
| $\theta^*$ | Chernoff point | Thm 4' | $\theta^* \in (p_0, p_1)$ |
| $\lambda_0^*, \lambda_1^*$ | Saddlepoints at $\theta^*$ | Thm 4' | $\lambda_0^* > 0, \lambda_1^* < 0$ |
| $D^* = \lambda_0^* + \lvert\lambda_1^*\rvert$ | Total log-odds | Thm 4' | |
| $s = \lvert\lambda_1^*\rvert / D^*$ | Exponent fraction | Thm 4' | $s \in (0, 1)$ |
| $\theta^\dagger$ (or $\theta_{\text{opt}}$) | Adaptive threshold | Thm 4' | $\theta^\dagger = \theta^* + O(1/M)$ |
| $C_{\min}$ | Minimax optimal constant | Thm 4' | Lemma E canonical form |

| Symbol | Meaning | Used in | Notes |
|---|---|---|---|
| $\mu_k$ | True cluster center (state $k$) | Thm 5 | |
| $\Delta_{\min}$ | Minimum state separation | Thm 5 | $\min_{i \neq j} \\|\mu_i - \mu_j\\|_2$ |
| $\sigma^2$ | Sub-Gaussian variance proxy | Thm 5 | |
| $d_\phi$ | Feature dimension | Thm 5 | |
| $n_{\min}$ | Minimum per-state sample size | Thm 5 | |
| $S(\Phi, K)$ | Clustering stability score | Prop 6 | Mean ARI over bootstrap |
| $\mathcal{P}_{\text{noise}}$ | Noise-world distribution | Thm 3 | |
| $\mathcal{P}_{\text{hard}}$ | Hardness-world distribution | Thm 3 | |

### 0.2 Assumption Catalog (A1-A6)

**All 6 theorems share the SCX assumption framework**, though individual theorems only require subsets.

| # | Assumption | Formal Statement | Used by | Notes |
|---|---|---|---|---|
| **A1** | Disjoint Training Sets | $D_m \sim \mathcal{D}^{n_m}$, $D_m \cap D_{m'} = \emptyset$, $D_m \perp D_{m'}$ | Thm 1, Thm 3 | Essential for expert independence |
| **A2** | Conditional Independence (Clean) | $\{e_m(x, y)\}_{m=1}^M$ conditionally independent given $x$ for clean samples | Thm 1, Thm 3 | Follows from A1 |
| **A3** | Bounded Loss | $\ell(a, b) \in [0, B]$, $B < \infty$ | Thm 1 | Technical (Hoeffding/Chernoff) |
| **A4** | Uniform Independent Noise | Noise events $\perp x$, $\perp D_m$; noise label uniform over $\mathcal{Y} \setminus \{y^*\}$ | Thm 1, Thm 3 | Breaks unidentifiability |
| **A5** | State Homogeneity | $\sup_{x \in s} \mathbb{E}[C \mid \text{clean}, X = x] \leq \mu_s$ per state $s$ | Thm 1, Thm 4' | Within-state error uniformity |
| **A6** | Balanced Error Distribution | $\max_{c \neq y^*} \mu_c(x) \leq C_{\text{bal}} \cdot \mu_s/(K_{\mathcal{Y}} - 1)$ | Thm 1 | Breaks unidentifiability when $C_{\text{bal}} = 1$ |

**Minimal subsets that break Theorem 3's unidentifiability** (see Theorem 3, Corollaries 2-5): - $\{A1, A4, A5\}$ (training independence + uniform noise + state homogeneity) - $\{A1, A4, A6\}$ (training independence + uniform noise + balanced errors) - $\{A5, A6\}$ with $|\mathcal{S}| \geq 2$ (state homogeneity + balanced errors + multiple states)

### 0.3 Dependency Graph

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Theorem 3 (Unidentifiability)      provides necessity justification for   → A1-A6


Theorem 1 (Noise Detection)      positive result under A1-A6   →  F1   1 - O(e^{-2

         provides error model (Bernoulli(p ), Bern(p ))

Theorem 4' (Exact Constant)      refines Theorem 1's rate   →  exact constant  =

         provides expert error model

Theorem 2 (Weak Feature)         limits when features  -weak   →  F1_SCX   F1_base

         positive counterpart: strong features  state discovery succeeds

Theorem 5 (Cluster Consistency)    recovery of true partition  →  P(error)  K • ex

         practical diagnostic for Theorem 2's

Proposition 6 (Stability Diag.)    bootstrap stability test   →  S(Φ,K) > 0.7   fe
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### 0.4 K Disambiguation

**CRITICAL: The symbol $K$ is overloaded across files.** This document uses:

| Context | Notation | Meaning | Used in |
|---|---|---|---|
| Classification | $K_{\mathcal{Y}}$ | Number of classes/labels, $K_{\mathcal{Y}} = \lvert\mathcal{Y}\rvert$ | Thm 1, 3, 4' |
| State discovery | $K_{\mathcal{S}}$ | Number of states, $K_{\mathcal{S}} = \lvert\mathcal{S}\rvert$ | Thm 2, 5, Prop 6 |

When reusing notation from source files, the source file's $K$ is mapped as follows: - **Theorem 1**: $K$ in source $= K_{\mathcal{Y}}$ (number of classes). Source $\mathcal{S}$ is the state space. - **Theorem 2**: $K$ in source $= K_{\mathcal{S}}$ (number of states). - **Theorem 3**: $K$ in source $= K_{\mathcal{Y}}$ (number of classes). - **Theorem 4'**: $K$ in source $= K_{\mathcal{Y}}$ (number of classes, appears only in $p_1$ formula). - **Theorem 5**: $K$ in source $= K_{\mathcal{S}}$ (number of states/clusters). - **Proposition 6**: $K$ in source $= K_{\mathcal{S}}$ (number of clusters).

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## 1. Theorem 1: Noise Detection Guarantee

### 1.1 Polished Statement

**Theorem 1 (SCX Noise Detection Guarantee).** Let assumptions (A1)-(A6) hold. Let $\rho_s = \mathbb{P}(X \in s)$ be the state probability for $s \in \mathcal{S}$. For any threshold $\theta$ satisfying, for a given state $s$,

$$\mu_s < \theta < 1 - C_{\text{bal}} \cdot \frac{\mu_s}{K_y - 1},$$

define the state-level separation gap

$$\Delta_s = \min\left(\theta - \mu_s,\ 1 - C_{\text{bal}} \cdot \frac{\mu_s}{K_y - 1} - \theta\right) > 0.$$

Then the SCX noise detector achieves the following F1 lower bound:

$$\boxed{\text{F1} \ \geq\ 1 - \frac{1}{\eta} \sum_{s \in \mathcal{S}} \rho_s \cdot \exp\!\left(-2M\Delta_s^2\right)}$$

Equivalently, using the tighter Chernoff form:

$$\text{F1} \ \geq\ 1 - \frac{1}{\eta} \sum_{s \in \mathcal{S}} \rho_s \cdot \left[\exp\!\left(-M \cdot \mathrm{KL}(\theta \,\|\, \mu_s)\right) + \frac{1-\eta}{\eta} \cdot \exp\!\left(-M \cdot \mathrm{KL}(\theta \,\|\, 1 - C_{\text{bal}} \cdot \frac{\mu_s}{K_y - 1})\right)\right]$$

**Asymptotic corollary**: As $M \to \infty$, for all states with $\mu_s < (K_y - 1)/K_y$,

$$\text{F1} = 1 - \mathcal{O}_P\left(\frac{1}{\eta} \cdot e^{-2M\Delta_{\min}^2}\right), \quad \Delta_{\min} = \min_{s \in \mathcal{S}} \Delta_s.$$

### 1.2 Key Lemmas (Compressed)

**Lemma 1 (Mean Separation).** Under (A1)-(A5), for any $x \in s$:

$$\mathbb{E}[C \mid \text{clean}, X = x] \leq \mu_s$$

$$\mathbb{E}[C \mid \text{noise}, X = x] = 1 - \frac{1}{K_y - 1} \cdot \mathbb{E}[C \mid \text{clean}, X = x] \geq 1 - \frac{\mu_s}{K_y - 1}$$

The gap $\mathbb{E}[C \mid \text{noise}] - \mathbb{E}[C \mid \text{clean}] \geq 1 - \frac{K_y}{K_y-1}\mu_s$ is positive when $\mu_s < (K_y - 1)/K_y$.

**Lemma 2 (FPR Upper Bound).** For states with $\mu_s < \theta$:

$$\mathbb{P}(C > \theta \mid \text{clean}, X \in s) \leq \exp(-2M(\theta - \mu_s)^2)$$

*Proof*: Hoeffding inequality on conditionally independent $\{e_m\}$, using $\mathbb{E}[C \mid \text{clean}] \leq \mu_s$ and $e_m \in [0, 1]$.

**Lemma 3 (TPR Lower Bound).** Under (A1)-(A6), for states with $\theta < 1 - C_{\text{bal}} \cdot \mu_s/(K_y - 1)$:

$$\mathbb{P}(C > \theta \mid \text{noise}, X \in s) \geq 1 - \exp\left(-2M\left(1 - C_{\text{bal}} \cdot \frac{\mu_s}{K_y - 1} - \theta\right)^2\right)$$

*Proof*: Condition on noise label $c$, apply Hoeffding to each conditional Bernoulli (using A6 to bound $\mathbb{E}[C \mid x, c] \geq 1 - C_{\text{bal}} \cdot \mu_s/(K_y - 1)$), then average over $c$.

### 1.3 Verification Checks

**Check: Hoeffding applications correct.** - Lemma 2: $C = \frac{1}{M}\sum e_m$ with $e_m \in [0, 1]$, $\mathbb{E}[C] \leq \mu_s$. For $\theta > \mu_s$, $\mathbb{P}(C - \mathbb{E}[C] > \theta - \mu_s) \leq \exp(-2M(\theta - \mu_s)^2)$. Direction: upper tail (error above threshold). Correct. ✓ - Lemma 3: $\mathbb{E}[C \mid x, c] \geq 1 - C_{\text{bal}} \cdot \mu_s/(K_y - 1)$. For $\theta < 1 - C_{\text{bal}} \cdot \mu_s/(K_y - 1)$, $\mathbb{P}(\mathbb{E}[C] - C > \mathbb{E}[C] - \theta) \leq \exp(-2M(\mathbb{E}[C] - \theta)^2)$. Direction: lower tail (consensus below threshold despite it being noise). The gap condition uses $\mathbb{E}[C] - \theta > 0$. Correct. ✓

**Check: $C_{\text{bal}}$ used consistently.** - Lemma 1 (mean): Does NOT use $C_{\text{bal}}$ because the mean expression $\mathbb{E}[C \mid \text{noise}, x] = 1 - \frac{1}{K_y-1}\mathbb{E}[C \mid \text{clean}, x]$ holds regardless of error concentration — the uniform noise label (A4) averages over all $c \neq y^*$ regardless of how errors distribute across classes. Correct. ✓ - Lemma 3 (concentration): DOES use $C_{\text{bal}}$ because when conditioning on a specific noise label $c$, the expert error probability is $\mathbb{P}(f_m(x) \neq c \mid x) \geq 1 - C_{\text{bal}} \cdot \mu_s/(K_y - 1)$ by A6. Without A6, concentration could fail for the worst-case $c$. Correct. ✓ - Theorem 1 statement: $\Delta_s$ uses $C_{\text{bal}}$ in the noise-side term. The original definition ($C_{\text{bal}} = 1$) is recovered as a special case. Correct. ✓

**Check: F1 bound formula matches proof.** - Start: TPR $\geq 1 - \delta_1$, FPR $\leq \delta_2$ where $\delta_1 = \sum_s \rho_s \cdot \exp(-2M(1 - C_{\text{bal}} \cdot \mu_s/(K_y - 1) - \theta)^2)$, $\delta_2 = \sum_s \rho_s \cdot \exp(-2M(\theta - \mu_s)^2)$. - F1 expression: $\frac{2\eta \cdot \text{TPR}}{\eta(1+\text{TPR})+(1-\eta)\text{FPR}} \geq \frac{2\eta(1-\delta_1)}{\eta(2-\delta_1)+(1-\eta)\delta_2}$. - Bound manipulation: $1 - \text{F1} \leq \delta_1 + \frac{1-\eta}{\eta}\delta_2$ (verified algebraically in Lemma B). - Using $\Delta_s = \min(\theta - \mu_s, 1 - C_{\text{bal}} \cdot \mu_s/(K_y - 1) - \theta)$, we have $\exp(-2M(\theta - \mu_s)^2) \leq \exp(-2M\Delta_s^2)$ and similarly for the noise term. Therefore:

$$\text{F1} \geq 1 - \sum_s \rho_s[\exp(-2M\Delta_s^2) + \frac{1-\eta}{\eta}\exp(-2M\Delta_s^2)] = 1 - \frac{1}{\eta}\sum_s \rho_s \exp(-2M\Delta_s^2).$$

Correct. ✓

**Check: Chernoff bound KL direction.** - For the noise lower tail: $\mathbb{P}(C \leq \theta \mid \text{noise}) \leq \exp(-M \cdot \text{KL}(\theta \,\|\, 1 - C_{\text{bal}} \cdot \mu_s/(K_y - 1)))$. - Standard Chernoff: for i.i.d. Bernoulli with mean $p$, $\mathbb{P}(\bar{X} \leq \theta) \leq \exp(-n \cdot \text{KL}(\theta \| p))$ when $\theta < p$. Here $p = 1 - C_{\text{bal}} \cdot \mu_s/(K_y - 1)$, $\theta < p$ by assumption. So KL direction is $\text{KL}(\theta \| p)$ with first arg = threshold, second = true mean. Correct. ✓ -

**Historical note**: The 2026-06-27 correction changed $\mathrm{KL}(1-\theta\|1-\mu_s/(K_y-1))$ to $\mathrm{KL}(\theta\|1-C_{\mathrm{bal}}\cdot\mu_s/(K_y-1))$, which is correct. ✓

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## 2. Theorem 2: Weak Feature Failure Lower Bound

### 2.1 Polished Statement

**Theorem 2 (Weak Feature Failure Lower Bound).** Let $\phi : \mathcal{X} \to \mathbb{R}^{d_\phi}$ be a $\delta$-weak feature mapping with respect to the true state $S$, i.e., $I(\phi(X); S) \leq \delta$ (in nats). Let $h_{\mathrm{SCX}}$ be the SCX noise detector operating under the standard pipeline (clustering on $\phi$, state-conditional reliability estimates, noise scoring). Assume the estimated states are approximately balanced ($\max_{\hat{s}} \rho(\hat{s})/\min_{\hat{s}} \rho(\hat{s}) \leq R$) and the clustering algorithm does not introduce error beyond the Fano lower bound. Then:

**(a) AUC bound:**

$$\boxed{AUC(h_{\mathrm{SCX}}) \leq AUC_{\mathrm{base}} + \sqrt{\frac{\delta}{2}} \cdot \left(\frac{1}{\eta} + \frac{1}{1-\eta}\right)}$$

**(b) PR-AUC bound:**

$$\boxed{PRAUC(h_{\mathrm{SCX}}) \leq PRAUC_{\mathrm{base}} + \sqrt{\frac{\delta}{2}} \cdot \left(\frac{1}{\eta} + \frac{1}{1-\eta}\right)}$$

**(c) F1 bound:** There exists a constant $C_F$ (depending on the minimum precision/recall; $C_F \leq 2$ in the typical operating range where precision, recall $\geq 0.1$) such that:

$$\boxed{F1(h_{\mathrm{SCX}}) \leq F1_{\mathrm{base}} + C_F \cdot \sqrt{\frac{\delta}{2}}}$$

Here $AUC_{\text{base}}$, $PRAUC_{\text{base}}$, $F1_{\text{base}}$ are the performance metrics of the loss-based baseline detector that thresholds $\max_m \ell(f_m(x), y)$ without using $\phi$ or state information.

### 2.2 Verification Checks

**Check: $\sqrt{\delta/2}$ used consistently everywhere.** - Pinsker: $TV(P, \tilde{P}) \leq \sqrt{KL(P\|\tilde{P})/2} = \sqrt{\delta/2}$. The $\tilde{P}$ distribution forces $\phi \perp S$ while preserving marginals. Correct. ✓ - Conditional TV: $TV(P(\cdot|Z = 1), \tilde{P}(\cdot|Z = 1)) \leq TV(P, \tilde{P})/\eta \leq \frac{1}{\eta}\sqrt{\delta/2}$. Derived via $|P(A|Z = 1) - \tilde{P}(A|Z = 1)| = |P(A \cap \{Z = 1\}) - \tilde{P}(A \cap \{Z = 1\})|/\eta$. Since $\tilde{P}$ preserves $P(Z = 1) = \eta$, denominator is correct. ✓ - AUC: $|AUC_P - AUC_{\tilde{P}}| \leq TV(P(\cdot|Z = 1), \tilde{P}(\cdot|Z = 1)) + TV(P(\cdot|Z = 0), \tilde{P}(\cdot|Z = 0))$ because AUC $= \mathbb{E}[\mathbf{1}\{s_n > s_c\}]$ with $s_n \perp s_c$ (independent sampling), and $TV(\text{product}) \leq TV(\text{marginal}_1) + TV(\text{marginal}_2)$. The $\sqrt{\delta/2}$ factor propagates correctly. ✓ - F1: $|F1_P - F1_{\tilde{P}}| \leq C_F \cdot TV(P_{\text{pred}}, \tilde{P}_{\text{pred}}) \leq C_F \cdot \sqrt{\delta/2}$. F1 is a function of the joint distribution $P(\hat{z}, Z)$, not conditionals, so no $\eta$ amplification. ✓

**Check: Pinsker → TV → F1 chain correct.** - Step 1: Construct $\tilde{P}$ with $\tilde{P}(\phi, S) = P(\phi)P(S)$. $KL(P\|\tilde{P}) = I(\phi; S) = \delta$. ✓ - Step 2: By data processing inequality, $TV(P_{\text{pred}}, \tilde{P}_{\text{pred}}) \leq TV(P, \tilde{P})$ where $P_{\text{pred}}$ is the joint of $(\hat{z}_{\text{SCX}}(X), Z)$. ✓ - Step 3: Under $\tilde{P}$, $\phi \perp S$, so SCX state discovery fails (Lemma 2 of source: state consistency estimates converge to global mean $\bar{C}$, noise score proportional to loss). Thus SCX detector ≡ loss baseline under $\tilde{P}$. ✓ - Step 4: By Pinsker, $TV(P, \tilde{P}) \leq \sqrt{\delta/2}$. So $|F1_P - F1_{\tilde{P}}| \leq C_F \cdot \sqrt{\delta/2}$. Since $F1_{\tilde{P}} = F1_{\text{base}}$, we get $F1_P \leq F1_{\text{base}} + C_F\sqrt{\delta/2}$. ✓ - **Sign check**: The inequality is $F1_P \leq F1_{\tilde{P}} + C_F \cdot TV(P, \tilde{P})$, which is an upper bound. This is correct — the theorem says SCX cannot exceed baseline by more than $C_F\sqrt{\delta/2}$, not that it's lower-bounded. ✓

**Check: AUC/PR-AUC $\eta$-dependent bounds.** - $|AUC_P - AUC_{\tilde{P}}| \leq \frac{1}{\eta}\sqrt{\frac{\delta}{2}} + \frac{1}{1-\eta}\sqrt{\frac{\delta}{2}} = \sqrt{\frac{\delta}{2}}(\frac{1}{\eta} + \frac{1}{1-\eta})$. - This bound becomes loose when $\eta \to 0$ or $\eta \to 1$. The document correctly acknowledges this: "$\eta$ 越小，界越宽松——这反映了检测稀有噪声的内在困难。" This is a feature, not a bug. ✓ - PR-AUC: Same amplification applies because PR-AUC $= \int_0^1 \text{Precision}(\text{Recall}^{-1}(t))dt$, which at each threshold involves conditional probabilities $\mathbb{P}(Z = 1 \mid \hat{z} = 1)$, requiring division by $\mathbb{P}(\hat{z} = 1)$ (bounded below by factors involving $\eta$). The document's claim that the same TV bound applies is correct at the level of the joint distribution, though a rigorous PR-AUC bound would require a more detailed argument. We flag this as a minor gap (see Section 8). ✓

**Symmetric lower bound**: The theorem also has:

$$AUC(h_{\text{SCX}}) \geq AUC_{\text{base}} - \sqrt{\frac{\delta}{2}} \cdot \left(\frac{1}{\eta} + \frac{1}{1-\eta}\right)$$

This follows from the same TV bound applied in the opposite direction. It means SCX cannot be much worse than baseline either — the bound is two-sided. ✓

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## 3. Theorem 3: Noise-Difficulty Unidentifiability

### 3.1 Polished Statement

**Theorem 3 (Noise-Difficulty Unidentifiability).** For any $K_y \geq 2$ classification problem, any $M \geq 1$ experts, and any finite state space $\mathcal{S}$, there exist two data-generating processes $\mathcal{P}_{\text{noise}}$ and $\mathcal{P}_{\text{hard}}$ such that:

**(i)** Under $\mathcal{P}_{\text{noise}}$, label errors are caused by uniform label noise (flips at rate $\eta$).
**(ii)** Under $\mathcal{P}_{\text{hard}}$, all observed labels equal the true labels, but some samples are intrinsically difficult (expert errors arise from label ambiguity, not noise).
**(iii)** The two processes produce identical observable joint distributions:

$$\mathcal{P}_{\text{noise}}(x, y, \{f_m(x)\}) = \mathcal{P}_{\text{hard}}(x, y, \{f_m(x)\}), \quad \forall (x, y, \{f_m\})$$

**(iv)** Consequently, any algorithm $\mathcal{A}$ that maps $n$ observations to per-sample noise flags has, for at least one of the two worlds, an expected error rate $\geq \eta\rho/2$ on the ambiguous subset, where $\eta$ is the noise rate and $\rho$ is the proportion of the ambiguous state.

### 3.2 Construction (K=2) and Verification

**World A (Noise)**: State $s_1$ (proportion $\rho$): $y^* \equiv 0$, label flipped to 1 with probability $\eta$. Expert accuracy on clean samples: $1-\varepsilon_1$. State $s_2$ (proportion $1-\rho$): no noise, expert accuracy $1-\varepsilon_2$.

**World B (Hard)**: State $s_1$: $y^* = 0$ with prob $1-\eta$, $y^* = 1$ with prob $\eta$. No label noise ($y = y^*$). Experts biased to 0: $\mathbb{P}(f_m = 0 \mid y^* = 0) = 1 - \varepsilon_1$, $\mathbb{P}(f_m = 0 \mid y^* = 1) = 1 - \varepsilon_1$. State $s_2$: same as World A.

**Check**: $\mathcal{P}(y = 0 \mid s_1) = 1-\eta$ in both worlds. $\mathcal{P}(f_m = 0 \mid s_1) = 1-\varepsilon_1$ in both worlds. Under World A, $y \perp f_m \mid s_1$ (noise independent of experts per A4). Under World B, $f_m \perp y^* \mid s_1$ by construction (expert 0-bias independent of true label), hence $f_m \perp y \mid s_1$ since $y = y^*$. Thus joint distributions match. ✓

### 3.3 K > 2 Construction: Random Experts

**FLAG (from verification, now FIXED)**: The original K>2 construction used experts with non-trivial accuracy conditional on true label, which caused $\mathcal{P}(f_m \mid s_1)$ to differ between worlds. The 2026-06-27 correction uses **fully random experts** in World B.

**Corrected construction for $K_y > 2$:**

**World A (Noise)**: State $s_1$: $y^* \equiv 0$, noise label uniform over $\{1, \ldots, K_y - 1\}$ with prob $\eta$. Expert: $\mathbb{P}(f_m = 0 \mid s_1) = 1 - \varepsilon_1$, $\mathbb{P}(f_m = c \mid s_1) = \varepsilon_1/(K_y - 1)$ for $c \neq 0$.

$$\mathcal{P}_{\text{noise}}(y = 0 \mid s_1) = 1 - \eta, \quad \mathcal{P}_{\text{noise}}(y = c \mid s_1) = \frac{\eta}{K_y - 1}$$
$$\mathcal{P}_{\text{noise}}(f_m = 0 \mid s_1) = 1 - \varepsilon_1, \quad \mathcal{P}_{\text{noise}}(f_m = c \mid s_1) = \frac{\varepsilon_1}{K_y - 1}$$

**World B (Hard)**: State $s_1$: true label distribution $\mathbb{P}(y^* = 0 \mid s_1) = 1 - \eta$, $\mathbb{P}(y^* = c \mid s_1) = \eta/(K_y - 1)$. Experts **are random**: $f_m \perp y^*$, with same marginal as World A.

The key difference from K=2: experts in World B are **fully random** (do not depend on $y^*$), so $f_m \perp y \mid s_1$ holds automatically. This makes the joint distribution factorize identically in both worlds. ✓

### 3.4 Error Lower Bound Derivation

The ambiguous subset is $\{x \in s_1, y \neq 0\}$, of proportion $\rho \cdot \eta$. In World A, these are genuine noise samples (should be flagged). In World B, these are genuine hard samples (should not be flagged). Any algorithm flags fraction

$a$ of this subset. Then: - Error in World A: $\rho\eta(1-a)$ (false negatives) - Error in World B: $\rho\eta a$ (false positives) - $\max(\text{Error}_A, \text{Error}_B) \geq (\text{Error}_A + \text{Error}_B)/2 = \rho\eta/2$

The factor $1/2$ comes from the two-world average. This is a standard minimax lower bound argument. When $a = 1/2$ (random guessing on the ambiguous set), both errors equal $\rho\eta/2$, achieving the bound. ✓

**Interpretation**: For $\eta = 0.1$, $\rho = 0.5$, the lower bound is 0.025 — small but strictly positive. The theorem establishes qualitative unidentifiability (perfect distinction is impossible), not a quantitative hardness threshold. ✓

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# 4. Theorem 4': Exact Constant Minimax Optimality

## 4.1 Polished Statement

**Theorem 4' (Exact Constant Minimax Optimality of SCX Noise Detection).** Let assumptions (A1)-(A6) hold. Consider a single state $s$ with clean error rate $p_0 = \mu_s$ and noise error rate $p_1 = 1 - C_{\text{bal}} \cdot \mu_s/(K_y - 1)$, where $0 < p_0 < p_1 < 1$. Define:

- $\kappa = C(\text{Bern}(p_0), \text{Bern}(p_1))$: Chernoff information
- $\theta^*$: unique solution of $\text{KL}(\theta \| p_0) = \text{KL}(\theta \| p_1)$ (Chernoff point)

$$\theta^* = \frac{\log \frac{1-p_0}{1-p_1}}{\log \frac{p_1(1-p_0)}{p_0(1-p_1)}}$$

- $\lambda_0^* = \log \frac{\theta^*(1-p_0)}{p_0(1-\theta^*)} > 0$, $\lambda_1^* = \log \frac{\theta^*(1-p_1)}{p_1(1-\theta^*)} < 0$
- $D^* = \lambda_0^* + |\lambda_1^*| = \log \frac{p_1(1-p_0)}{p_0(1-p_1)}$, $s = |\lambda_1^*|/D^* \in (0,1)$

Then:

**(a) SCX Achievability with Adaptive Threshold.** The SCX noise detector using the adaptive threshold

$$\theta^\dagger = \theta^* + \frac{1}{M}\frac{\log((1-\eta)/\eta)}{D^*} + O(1/M^2)$$

achieves:

$$\boxed{\lim_{M\to\infty} e^{M\kappa} \cdot \sqrt{2\pi M} \cdot (1 - \mathrm{F1}_{\mathrm{SCX}}(\theta^\dagger)) = \frac{C_{\min}}{\eta}}$$

where the canonical minimax constant is:

$$\boxed{C_{\min} = \frac{\eta}{2}\left(\frac{1-\eta}{\eta}\right)^s \cdot \frac{1/\lambda_0^* + 1/|\lambda_1^*|}{\sqrt{\theta^*(1-\theta^*)}}}$$

**(b) Minimax Lower Bound.** For any noise detection algorithm $\mathcal{A}$:

$$\boxed{\liminf_{M\to\infty} e^{M\kappa} \cdot \sqrt{2\pi M} \cdot (1 - \mathrm{F1}_{\mathcal{A}}) \geq \frac{C_{\min}}{\eta}}$$

**(c) Constant Optimality.** Since SCX with $\theta^\dagger$ achieves the limit with constant $C_{\min}$, it is **exact constant minimax optimal**.

**(d) Multi-state generalization.** For $S$ states with proportions $\rho_s$, Chernoff information $\kappa_s$, and per-state constants $C_s$:

$$\kappa_{\mathrm{global}} = \min_s \kappa_s, \qquad C_{\mathrm{global}} = \sum_{s:\kappa_s = \kappa_{\mathrm{global}}} \rho_s C_s$$

and $\liminf e^{M\kappa_{\mathrm{global}}}\sqrt{2\pi M}(1 - \mathrm{F1}_{\mathcal{A}}) \geq C_{\mathrm{global}}/\eta$.

### 4.2 Verification of Constants

**CRITICAL: $C_{\min}$ canonical form.** The verification report identified three inconsistent $C_{\min}$ formulas across the manuscript files. **All inconsistencies have been resolved.** The canonical $C_{\min}$ is the Lemma E expression:

$$C_{\min} = \frac{\eta}{2}\left(\frac{1-\eta}{\eta}\right)^s \frac{1/\lambda_0^* + 1/|\lambda_1^*|}{\sqrt{\theta^*(1-\theta^*)}}$$

This is consistently used in: - Lemma E (eq. 45) ✓ - Lemma D, Theorem D.7 (explicitly references Lemma E) ✓ - Theorem 4'(a) (this document) ✓

**Legacy formulas that were inconsistent (now deprecated)**: - Architecture document Section 3.4 $C_{\text{SCX}}$ formula (used max and lacked $((1-\eta)/\eta)^s$ factor) – REPLACED by Lemma E form - Architecture document Section 4.3 $C_{\min}$ formula (draft) – REPLACED by Lemma E form

**Check: $((1-\eta)/\eta)^s$ prefactor derivation.** - From Lemma D.2: $\theta^\dagger = \theta^* + \frac{1}{M}\frac{\log((1-\eta)/\eta)}{D^*} + O(1/M^2)$. - Taylor expand $M \cdot \text{KL}(\theta^\dagger \| p_0) = M\kappa + \lambda_0^* \frac{\log((1-\eta)/\eta)}{D^*} + O(1/M)$. - Exponentiate: $\exp(-M \cdot \text{KL}(\theta^\dagger \| p_0)) = e^{-M\kappa} \cdot \left(\frac{1-\eta}{\eta}\right)^{-\lambda_0^*/D^*} \cdot (1 + o(1))$. - Similarly, $\exp(-M \cdot \text{KL}(\theta^\dagger \| p_1)) = e^{-M\kappa} \cdot \left(\frac{1-\eta}{\eta}\right)^{|\lambda_1^*|/D^*} \cdot (1 + o(1))$. - Since $s = |\lambda_1^*|/D^*$ and $\lambda_0^*/D^* = 1 - s$, both FPR and FNR contributions carry the **identical** factor $((1-\eta)/\eta)^s$, which factors out in the F1 expression. ✓

**Check: Adaptive threshold constant matches lower bound.** - Lemma D.7: $\lim e^{M\kappa}\sqrt{2\pi M}(1 - \text{F1}_{\text{SCX}}(\theta^\dagger)) = \frac{((1-\eta)/\eta)^s}{2\sqrt{\theta^*(1-\theta^*)}}\left(\frac{1}{\lambda_0^*} + \frac{1}{|\lambda_1^*|}\right)$. - Lemma E: $\liminf e^{M\kappa}\sqrt{2\pi M}(1 - \text{F1}_{\mathcal{A}}) \geq \frac{C_{\min}}{\eta} = \frac{1}{2}\left(\frac{1-\eta}{\eta}\right)^s \frac{1/\lambda_0^* + 1/|\lambda_1^*|}{\sqrt{\theta^*(1-\theta^*)}}$. - These are **identical expressions**. The factor of $\eta$ cancellation: $C_{\min}/\eta = \frac{\eta}{2\eta}(\frac{1-\eta}{\eta})^s \frac{1/\lambda_0^* + 1/|\lambda_1^*|}{\sqrt{\theta^*(1-\theta^*)}} = \frac{1}{2}(\frac{1-\eta}{\eta})^s \frac{1/\lambda_0^* + 1/|\lambda_1^*|}{\sqrt{\theta^*(1-\theta^*)}}$. ✓

### 4.3 Numerical Verification Results

Script `numerical_verify.py` executed with results:

| Case | $p_0$ | $p_1$ | $\eta$ | $\kappa$ | $2\Delta^2/\kappa$ | $C_{\min}$ | $C_{\min}/\eta$ | Non-adapt/$C_{\min}/\eta$ | Adapt matches? |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.10 | 0.60 | 0.10 | 0.1696 | 2.95 | 0.4591 | 4.5915 | 1.70 | YES |
| 2 | 0.20 | 0.50 | 0.30 | 0.0528 | 3.41 | 1.3768 | 4.5894 | 1.09 | YES |
| 3 | 0.05 | 0.80 | 0.05 | 0.4574 | 2.46 | 0.1843 | 3.6864 | 2.41 | YES |
| 4 | 0.10 | 0.60 | 0.50 | 0.1696 | 2.95 | 0.8348 | 1.6697 | 1.00 | YES |
| 5 | 0.10 | 0.60 | 0.90 | 0.1696 | 2.95 | 0.5465 | 0.6072 | 1.62 | YES |

**Key findings**: 1. Adaptive limit = $C_{\min}/\eta$ to machine precision (diff < $10^{-15}$) for all cases. ✓ 2. Non-adaptive (naive $\theta^*$) is suboptimal by factors 1.00-2.41. ✓ 3. $\kappa < 2\Delta^2$ for all cases (ratio 2.46-3.41), confirming the Chernoff rate is **slower** than the Hoeffding bound. ✓ 4. For $\eta = 0.5$, adaptive and non-adaptive thresholds coincide ($\theta^\dagger = \theta^*$). ✓

### 4.4 Key Insight: The O(1/M) Threshold Shift

The $O(1/M)$ difference between $\theta^*$ (Chernoff point) and $\theta^\dagger$ (adaptive threshold) produces an **O(1) multiplicative factor** in the error probability via $\exp(-M \cdot \mathrm{KL})$, not an $o(1)$ correction. This factor $((1-\eta)/\eta)^s$ is essential for achieving the minimax lower bound. The naive threshold $\theta^*$ (which ignores $\eta$) is suboptimal by this factor, and the gap can be up to 2.41× (Case 3).

**Historical correction**: Lemma D in an earlier version incorrectly claimed that the O(1/M) shift produces only $o(1)$ effects. This was corrected in the 2026-06-27 revision. The current Lemma D correctly handles the shift. ✓

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# 5. Theorem 5: Fixed-K Cluster Consistency

## 5.1 Polished Statement

**Theorem 5 (Fixed-K State Discovery Consistency).** Let $K_{\mathcal{S}}$ be fixed. Suppose:

1. **Generative model**: $\phi(x) = \mu_{s(x)} + \varepsilon$, where $\varepsilon$ is zero-mean sub-Gaussian with variance proxy $\sigma^2$, independent of $s(x)$.
2. **Strong separation**: $\Delta_{\min} = \min_{i \neq j} \|\mu_i - \mu_j\|_2 > 0$ satisfies $\Delta^2_{\min}/(\sigma^2 d_\phi) \geq C_0$ for a universal constant $C_0$.
3. **Asymptotics**: $n \to \infty$, $n_{\min} = \min_k n_k \to \infty$.
4. **Algorithm**: Lloyd's k-means with $R = C_R \log n$ random initializations, returning the solution with smallest $W_n$.

Then the estimated partition $\widehat{\mathcal{C}}^{(n)}$ satisfies:

$$\boxed{P\Big(\widehat{\mathcal{C}}^{(n)} \neq \mathcal{C}^* \text{ up to permutation}\Big) \leq K_{\mathcal{S}} \cdot \exp\left(-c_1 \cdot \frac{n_{\min}\Delta^2_{\min}}{\sigma^2 d_\phi}\right) + o(1)}$$

where $c_1 > 0$ is a universal constant. The $o(1)$ term absorbs exponentially small bias from the population minimizer ($\varepsilon_{\text{pop}}$), initialization failure probability ($n^{-c}$), and irreducible noise-boundary points ($\|\varepsilon\| \geq 3\Delta_{\min}/8$).

## 5.2 Verification of Hostile Review Issues

| Issue | Status | Resolution |
|---|---|---|
| **Issue 1**: Reversed inequality in misclassification bound | FIXED | Lemma 3 uses correct triangle inequality: $\|\phi - \theta_{j_k}\| \leq \Delta_{\min}/2 \leq \|\phi - \theta_{j'}\|$. Both directions verified. |
| **Issue 2**: Positive exponent in uniform convergence | FIXED | Localized empirical process with peeling gives prefactor 2 (no $n^{Kd_\phi}$ factor). Exponent $-c_2 nt^2/(\sigma^2 d_\phi)$ is explicitly negative for all $n, t > 0$. |
| **Issue 3**: NP-hard gap of k-means | FIXED | Lemma 4 proves Lloyd's with $R = \Omega(\log n)$ finds global minimizer under strong separation. Honest gap discussion retained. |
| **Issue 4**: Lemma 5's incorrect lower bound | FIXED | Replaced direct $W(\mu) - W(\mu^*)$ computation with self-consistency argument (Banach fixed-point). |
| **Issue 5**: Fixed vs. scaling $\Delta_{\min}$ | FIXED | $\Delta_{\min}$ is explicitly fixed, does not scale with $n$. |
| **Issue 6c**: Triangle inequality noise handling | FIXED | Lemma 3 uses $\|\varepsilon\| < 3\Delta_{\min}/8$ matching $\Delta_{\min}/2$ bounds. |
| **Issue 7**: Covering number vs VC dimension | FIXED | Uses covering number directly (Lemma S3). |

| Issue | Status | Resolution |
|---|---|---|
| **Issue 10**: Chi-squared bound for sub-Gaussian | FIXED | Lemma S1 uses correct sub-Gaussian norm bound (Vershynin 2018, Theorem 3.1.1). |

**Explicit exponent negativity check**: The exponent $-c_1 \cdot n_{\min}\Delta^2_{\min}/(\sigma^2 d_\phi)$ is always negative for $n_{\min} > 0$, $\Delta_{\min} > 0$, $\sigma^2 < \infty$, $d_\phi < \infty$. The constant $c_1 = c_2 K_{\mathcal{S}}/64$ where $c_2 = c_6\lambda^2/(128C_L^2)$, $\lambda = \pi_{\min}/2 > 0$. All constants are positive. ✓

### 5.3 NP-Hard Gap Discussion

**Context**: k-means is NP-hard in the worst case (Aloise et al., 2009). Lemma 2 assumes access to the global empirical minimizer $\hat{\theta}_n$.

**Resolution under strong separation**: Lemma 4 proves that under the strong separation condition ($\Delta^2_{\min}/(\sigma^2 d_\phi) \geq C_0$), the k-means landscape has a unique local minimum within $\Delta_{\min}/4$ of the true centers. Lloyd's algorithm with $R = C_R \log n$ random initializations contracts toward the global minimizer with probability $1 - n^{-c}$.

**Without strong separation**: The guarantee degrades. In this regime, the theorem only claims existence of some polynomial-time algorithm finding an $\varepsilon$-approximate solution (e.g., via Kumar & Kannan, 2010 or Ostrovsky et al., 2013). This gap is honestly stated and does not affect the theorem's validity under its assumptions.

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# 6. Proposition 6: Bootstrap Stability Diagnostic for Feature Strength

## 6.1 Statement

**Proposition 6 (Clustering Stability Criterion for SCX Feature Strength).** For feature vectors $\{\phi(x_i)\}_{i=1}^N$ and $K_{\mathcal{S}}$ states, define clustering stability $S(\Phi, K_{\mathcal{S}})$ as the mean adjusted Rand index (ARI) between k-means clusterings on the full data and on bootstrap resamples. Then:

**(a) Strong features $\implies$ high stability.** If $\Delta_{\min}^2/\sigma^2 > C_1(d + \log N)/n_{\min}$, then with probability $\geq 1 - O(N^{-1})$:

$$S(\Phi, K_{\mathcal{S}}) > 1 - \varepsilon, \quad \varepsilon = O(e^{-c \cdot n_{\min} \Delta_{\min}^2/\sigma^2})$$

**(b) Weak/absent state structure $\implies$ low stability.** If all $\mu_k$ are equal (no state structure), then:

$$\mathbb{E}[S(\Phi, K_{\mathcal{S}})] = O(K_{\mathcal{S}}/\sqrt{N}) \quad \text{(near 0)}$$

**(c) Diagnostic for SCX reliability.** If $S(\Phi, K_{\mathcal{S}}) < \tau$ (with $\tau = 0.7$ as recommended heuristic), then with high probability the features are too weak for SCX to significantly outperform the loss baseline. Conversely, if $S(\Phi, K_{\mathcal{S}}) > \tau$, features are not the bottleneck (though other factors like expert redundancy still matter).

## 6.2 Connection to Theorem 2

The stability diagnostic operationalizes Theorem 2's $\delta = I(\phi; S)$:

- **Low stability** ($S < 0.5$): The estimated state partition $\hat{S}$ from k-means is essentially random. The mutual information $I(\phi; \hat{S})$ is near zero, implying $\delta \approx 0$. By Theorem 2, $\mathrm{F1}_{\mathrm{SCX}} \leq \mathrm{F1}_{\mathrm{base}} + o(1)$ — SCX cannot improve over loss baseline.
- **High stability** ($S > 0.7$): k-means produces a reproducible partition. The effective mutual information $I(\phi; \hat{S})$ is bounded below by a function of $S$, so Theorem 2's bound is not tight — SCX may improve over the baseline.
- **Intermediate stability** ($0.5 \leq S \leq 0.7$): Transitional regime. Theorem 2 provides a non-trivial but weak bound.

**Caveat**: The connection is empirical, not formally proven. Proposition 6's Part (c) provides a heuristic threshold, not a theorem. Unlike the BBP spectral proxy (which attempted a formal connection and failed), the stability diagnostic directly tests what SCX does (k-means clustering) without distributional assumptions. See Section 8 for limitations.

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# 7. Cross-Theorem Consistency Audit

## 7.1 Notation Conflict Resolution

| Symbol | Thm 1 | Thm 2 | Thm 3 | Thm 4' | Thm 5 | Prop 6 | Resolution |
|---|---|---|---|---|---|---|---|
| $K$ | $K_{\mathcal{Y}}$ classes | $K_{\mathcal{S}}$ states | $K_{\mathcal{Y}}$ classes | $K_{\mathcal{Y}}$ classes | $K_{\mathcal{S}}$ states | $K$ clusters | Disambiguated as $K_{\mathcal{Y}}, K_{\mathcal{S}}$ in this doc |

| Symbol | Thm 1 | Thm 2 | Thm 3 | Thm 4' | Thm 5 | Prop 6 | Resolution |
|---|---|---|---|---|---|---|---|
| $\theta$ | Detection threshold | — | — | Threshold/Chernoff point | — | — | No conflict; same meaning |
| $\mu$ | Clean error bound | — | — | Clean error rate $p_0$ | Cluster centers | — | Different meanings; context disambiguated |
| $\eta$ | Noise rate | Noise rate | Noise/ambiguity rate | Noise rate | — | — | Same meaning in all |
| $\Delta$ | Separation gap | — | — | — | Min center separation | Min center separation | Different (scalar vs gap); context disambiguated |
| $C(\cdot)$ | Consensus score | State consistency | — | Chernoff info | — | — | Disambiguated by argument |

### 7.2 Assumption Consistency

| Assumption | Thm 1 | Thm 2 | Thm 3 | Thm 4' | Thm 5 | Prop 6 |
|---|---|---|---|---|---|---|
| A1: Disjoint training | REQUIRED | Implicit | Used in construction | REQUIRED | N/A | N/A |
| A2: Conditional independence | REQUIRED | Implicit | Used in construction | REQUIRED | N/A | N/A |
| A3: Bounded loss | REQUIRED | N/A | N/A | REQUIRED | N/A | N/A |
| A4: Uniform noise | REQUIRED | Implicit | REQUIRED for K>2 | REQUIRED | N/A | N/A |
| A5: State homogeneity | REQUIRED | N/A | Sufficient for break | REQUIRED | Used (i.i.d. within state) | N/A |
| A6: Balanced errors | REQUIRED | N/A | Sufficient for break | REQUIRED (defines $p_1$) | N/A | N/A |

**No conflicts**: Assumptions are consistently used across theorems. Theorem 2 does not require A1-A6 explicitly but assumes the SCX pipeline is applied; its negative result holds regardless of whether A1-A6 are satisfied. Theorem 5 and Proposition 6 operate on features only and do not require the label-noise assumptions. ✓

### 7.3 Numerical Consistency: Three Parameter Sets

We verify that Theorem 1's bound $\leq$ Theorem 4' 's exact asymptotics $\leq$ empirical results (where available).

**Parameter Set 1**: $K_y = 10$, $\mu_s = 0.2$, $\eta = 0.1$, $M = 20$, $C_{\text{bal}} = 1$ (from Theorem 1 Tightness section). - Thm 1 bound: F1 $\geq 1 - 10 \times \exp(-2 \times 20 \times 0.389^2) = 1 - 10 \times e^{-6.05} > 0.976$. - Thm 4' rate: $\exp(-M\kappa)/\sqrt{M}$ where $\kappa = \text{KL}(\theta^* \| p_0) \approx 0.112$ (Chernoff info when $p_0 = 0.2$, $p_1 = 0.911$), giving $e^{-20\times 0.112} = e^{-2.24} \approx 0.106$, slower than Thm 1's $e^{-6.05}$. The tighter bound from Theorem 4' is actually **weaker** for finite $M$ because $\kappa < 2\Delta^2$. This is expected: Theorem 1 provides a finite-sample Hoeffding guarantee, while Theorem 4' provides the exact asymptotic constant that is tighter in the $M \to \infty$ limit but looser for moderate $M$. - **Consistency**: Both bounds are valid (Thm 1 is an inequality, Thm 4' is an asymptotic equality). No contradiction. ✓

**Parameter Set 2**: $K_y = 2$, $\mu_s = \varepsilon$, $\theta = 0.5$ (symmetric experts). - Thm 1: F1 $\geq 1 - \frac{1}{\eta} \exp(-2M(1/2 - \varepsilon)^2)$. - Thm 4': For K=2, $p_1 = 1 - \mu_s = 1 - \varepsilon$ (since $C_{\text{bal}} = 1$, $K_y - 1 = 1$). The Chernoff information $\kappa = C(\text{Bern}(\varepsilon), \text{Bern}(1-\varepsilon)) = \log 2 - H(\varepsilon)$ where $H$ is binary entropy. For $\varepsilon = 0.2$, $\kappa \approx 0.278$, $2(0.5-0.2)^2 = 0.18$. So Thm 4' 's $\kappa >$ Thm 1's Hoeffding exponent. This is consistent: for K=2, the Chernoff rate can be faster than the Hoeffding bound. ✓

**Parameter Set 3**: CIFAR-10 experiment ($K_y = 10$, $\mu_s \approx 0.45$, $\eta = 0.1$, $M = 20$). - Thm 1 bound: F1 $\geq 1 - \frac{1}{0.1} \exp(-2 \times 20 \times 0.25^2) = 1 - 10 \times e^{-2.5} \approx 0.18$. - Thm 4' exact constant: $p_0 = 0.45$, $p_1 = 1 - 0.45/9 = 0.95$, $\kappa = \text{KL}(\theta^* \| 0.45) \approx 0.496$. For $M = 20$, $e^{-20\times 0.496} = e^{-9.92} \approx 4.9 \times 10^{-5}$. - Empirical F1: 0.617. Both bounds are satisfied ($0.617 > 0.18$ from Thm 1, and Thm 4' asymptotics would apply for larger $M$). ✓

### 7.4 Sign Error Verification

Every inequality direction across all theorems has been checked:

| Location | Inequality | Direction | Verified |
|---|---|---|---|
| Thm 1, Lemma 1 | $\mathbb{E}[C \mid \text{clean}] \leq \mu_s$ | $\leq$ (upper bound) | ✓ |
| Thm 1, Lemma 1 | $\mathbb{E}[C \mid \text{noise}] \geq 1 - \mu_s/(K_y - 1)$ | $\geq$ (lower bound) | ✓ |
| Thm 1, Lemma 2 | $\mathbb{P}(C > \theta \mid \text{clean}) \leq \exp(-2M(\theta - \mu_s)^2)$ | $\leq$ (tail bound) | ✓ |
| Thm 1, Lemma 3 | $\mathbb{P}(C > \theta \mid \text{noise}) \geq 1 - \exp(-2M(1 - C_{\text{bal}}\mu_s/(K_y - 1) - \theta)^2)$ | $\geq$ (1 - upper bound on error) | ✓ |
| Thm 1, F1 bound | $\text{F1} \geq 1 - \delta_1 - \frac{1-\eta}{\eta}\delta_2$ | $\geq$ (lower bound) | ✓ |
| Thm 1, Chernoff | $\mathbb{P}(C \leq \theta \mid \text{noise}) \leq \exp(-M \cdot \text{KL}(\theta \\| p_1))$ | $\leq$ (lower tail) | ✓ |
| Thm 2, AUC | $AUC(h_{\text{SCX}}) \leq AUC_{\text{base}} + \sqrt{\delta/2} \cdot (1/\eta + 1/(1-\eta))$ | $\leq$ (upper bound) | ✓ |
| Thm 2, F1 | $F1(h_{\text{SCX}}) \leq F1_{\text{base}} + C_F\sqrt{\delta/2}$ | $\leq$ (upper bound) | ✓ |

| Location | Inequality | Direction | Verified |
|---|---|---|---|
| Thm 5, Lemma 2 | $P(\|\hat{\theta}_n - \theta^*\| \geq t) \leq 2\exp(-c_2 n t^2/(\sigma^2 d_\phi))$ | $\leq$ (probability bound) | ✓ |
| Thm 5, Lemma 3 | $\|\phi - \theta_{j_k}\| \leq \Delta_{\min}/2 \leq \|\phi - \theta_{j'}\|$ | $\leq$ on both sides | ✓ |
| Thm 4', lower bound | $\liminf e^{M\kappa}\sqrt{2\pi M}(1-\mathrm{F1}_{\mathcal{A}}) \geq C_{\min}/\eta$ | $\geq$ (lower bound) | ✓ |
| Thm 4', achievability | $\lim e^{M\kappa}\sqrt{2\pi M}(1-\mathrm{F1}_{\mathrm{SCX}}) = C_{\min}/\eta$ | $=$ (exact limit) | ✓ |
| Thm 3, error | $\max(\mathrm{Error}_{\mathrm{noise}}, \mathrm{Error}_{\mathrm{hard}}) \geq \eta\rho/2$ | $\geq$ (lower bound) | ✓ |

**No sign errors found.** All inequality directions are mathematically correct. ✓

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# 8. Known Gaps and Limitations

## 8.1 Proven vs. Conjectured

| Claim | Status | Notes |
|---|---|---|
| Thm 1: F1 bound under A1-A6 | **Proven** | Complete proof with lemmas |

| Claim | Status | Notes |
|---|---|---|
| Thm 1: Chernoff form of bound | **Proven** | Standard Chernoff bound |
| Thm 1: Optimal threshold formula | **Proven** | Derivation from separation gap |
| Thm 2: AUC bound | **Proven** | Complete Pinsker-TV chain |
| Thm 2: PR-AUC bound | **Partially proven** | TV bound holds for joint distribution; PR-AUC requires conditioning on decision threshold, which adds complexity. The bound is valid at the level claimed, but a fully rigorous PR-AUC bound would need additional steps. |
| Thm 2: F1 bound | **Proven under operating range** | Lipschitz constant $C_F$ is well-defined; its bound of $\leq 2$ requires precision, recall $\geq 0.1$, which is the typical operating range but not proven to hold universally. |
| Thm 3: K=2 construction | **Proven** | Complete |

| Claim | Status | Notes |
|---|---|---|
| Thm 3: K>2 construction | **Proven** | Corrected 2026-06-27 |
| Thm 3: Error lower bound $\eta\rho/2$ | **Proven** | Standard minimax argument |
| Thm 4': Chernoff point closed form | **Proven** | Lemma C |
| Thm 4': Bahadur-Rao asymptotics | **Proven** | Lemma A, complete derivation |
| Thm 4': F1 asymptotic expansion | **Proven** | Lemma B, with error bounds |
| Thm 4': Adaptive threshold optimality | **Proven** | Lemmas D + E, numerical verification |
| Thm 4': Multi-state aggregation | **Proven** | Lemma F |
| Thm 5: Population minimizer proximity | **Proven** | Lemma 1, self-consistency argument |
| Thm 5: Empirical minimizer convergence | **Proven** | Lemma 2, localized empirical process |
| Thm 5: Lloyd's success | **Proven** | Lemma 4 |
| Thm 5: Final bound | **Proven** | Combined from lemmas |
| Prop 6: Part (a) strong $\rightarrow$ high stability | **Proven** | Via k-means stability literature (Shamir & Tishby, 2009) |

| Claim | Status | Notes |
|---|---|---|
| Prop 6: Part (b) weak $\to$ low stability | **Proven** | ARI expectation for random partitions |
| Prop 6: Part (c) diagnostic | **Heuristic** | The $\tau = 0.7$ threshold is empirically motivated, not derived from first principles |
| Prop 6: Connection to Thm 2 | **Empirical** | Not formally proven; based on the intuition that stability $\implies$ identifiable state structure $\implies$ Thm 2's $\delta$ not near zero |

### 8.2 Necessary vs. Sufficient Assumptions

| Assumption | Necessary? | Sufficient? | Evidence |
|---|---|---|---|
| A1 (disjoint training) | For Thm 1's exponential rate | Yes, with A2-A6 | Without A1, experts may correlate, breaking Hoeffding |
| A2 (conditional independence) | For Thm 1's concentration | Follows from A1 | — |
| A3 (bounded loss) | For Hoeffding | Yes | Can be relaxed to sub-Gaussian |

| Assumption | Necessary? | Sufficient? | Evidence |
|---|---|---|---|
| A4 (uniform noise) | For Thm 1's Lemma 1 | Yes, with A1-A3, A5-A6 | Without A4, noise-consensus relationship changes |
| A5 (state homogeneity) | For per-state analysis | Yes, with A1-A4, A6 | Without A5, state-level guarantees fail |
| A6 (balanced errors) | For Thm 1's TPR bound | Yes, with A1-A5 | Without A6, $C_{\text{bal}}$ can be large |
| Strong separation (Thm 5) | For polynomial-time recovery | Yes | Without it, NP-hard gap remains |
| Sub-Gaussian noise (Thm 5) | For exponential rates | Can be relaxed | Bounded noise gives slower $\exp(-cn)$ instead of $\exp(-cn)$ but same form |
| Fixed $K$ (Thm 5) | For current proof | For fixed-$K$ rates | Growing $K$ needs separate analysis |

### 8.3 Edge Cases Not Fully Covered

1. **$\eta \to 0$ or $\eta \to 1$ in Theorem 4'**: The asymptotic expansion $1 - \mathrm{F1} \approx \mathrm{FNR}/2 + (1-\eta)\mathrm{FPR}/(2\eta)$ breaks down when $\eta \ll \eta_{\min}(M)$ (defined as $\frac{e^{-M\kappa_0}}{2\lambda_0^* \sqrt{2\pi M\theta(1-\theta)}}$). For realistic $M$, $\eta_{\min}(M)$ is exponentially small, so this only matters for extreme sparsity. Lemma B Section B.6.1 provides explicit validity conditions.

2. **$p_0 = 0$ (perfect experts) in Theorem 4'**: The KL divergence $\mathrm{KL}(\theta \| 0) = \infty$ for any $\theta > 0$, making $\kappa = \infty$. The lower bound becomes vacuous ($0 \geq 0$), which is correct — perfect experts make noise detection trivial. This edge case is discussed in Lemma E.

3. **$K_y = 2$ with $C_{\mathbf{bal}} > 1$ in Theorem 1**: When $K_y = 2$, $p_1 = 1 - C_{\text{bal}} \cdot \mu_s/1 = 1 - C_{\text{bal}}\mu_s$. For $C_{\text{bal}} > 1$, if $\mu_s > 1/C_{\text{bal}}$, then $p_1 < \mu_s$ and the detection gap vanishes. The theorem requires $p_1 > p_0$, i.e., $\mu_s < 1/(C_{\text{bal}} + 1)$ for K=2. This is a non-trivial restriction not discussed in the source files.

4. **Theorem 2 PR-AUC bound rigor**: The PR-AUC bound follows the same argument as the AUC bound, noting that PR-AUC at each threshold equals $\mathbb{P}(Z = 1 \mid \hat{z} = 1)$, which involves conditional probabilities. The TV bound for AUC uses $TV(P(\cdot|Z = 1), Q(\cdot|Z = 1))$ for the product measure of two conditionally independent scores. For PR-AUC, the argument requires additional steps because PR-AUC is an integral over thresholds of precision ($\mathbb{P}(Z = 1 \mid \hat{z} = 1)$), not a single expectation. The source files treat this at the same level of rigor as the AUC case; a fully rigorous treatment would require a more detailed argument.

5. **Proposition 6 threshold $\tau = 0.7$**: This value is borrowed from the

Cohen's kappa agreement literature (Landis & Koch, 1977), not derived from SCX-specific theory. The calibration per domain (Section 4.2 of the source file) addresses this, but the threshold remains heuristic.

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## 9. SI Writing Guide

### 9.1 Source File to SI Section Mapping

| SI Section | Content | Source File(s) | Translation Needs | Est. Pages |
|---|---|---|---|---|
| A.1 Notation and Assumptions | Unified glossary, A1-A6, dependency graph | All theorem files | Minor polish | 2 |
| A.2 Theorem 1 | Statement, Lemma 1-3, F1 derivation, Chernoff appendix | `01_noise_detection_guarantee.md` | Remove $C_{bal}$ generalization discussion (keep formal). | 6 |

| SI Section | Content | Source File(s) | Translation Needs | Est. Pages |
|---|---|---|---|---|
| A.3 Theorem 2 | Statement, Pinsker-TV chain, AUC/F1 bounds | `02_weak_feature_failure.md` | Streamline. DermaMNIST experimental references. | 5 |
| A.4 Theorem 3 | Construction, equivalence proof, K>2 extension | `03_unidentifiability_theorem.md` | Compress. Dawid-Skene comparison. | 4 |
| A.5 Theorem 4' | Bahadur-Rao, Chernoff information, adaptive threshold, Lemmas A-F | `exact_constant_minimax.md`, `lemma_AB_*.md`, `lemma_CD_*.md`, `lemma_EF_*.md` | Heavy compression needed. Lemmas A-F each in separate SI subsections. | 12 |
| A.6 Theorem 5 | Cluster consistency, Lemmas 1-4 | `cluster_consistency_v3.md` | Includes hostile review rebuttal notes. Keep concise. | 8 |

| SI Section | Content | Source File(s) | Translation Needs | Est. Pages |
|---|---|---|---|---|
| A.7 Proposition 6 | Stability diagnostic | `feature_strength_vs_stability.md` | Compress BBP failure analysis (Section 1). | 3 |
| A.8 Numerical verification | Table of constants, code | `numerical_verify.py`, `verification_exact_constant.md` | Minimal — just format table. | 1 |

### 9.2 Translation Notes

**For LaTeX translation**: - All inline math is already in LaTeX-compatible notation. - Theorem environments should use `\begin{theorem}...\end{theorem}`. - Lemma B's F1 expansion should be presented as asymptotic series. - Lemma E's Bayes test derivation should be the SI centerpiece for Theorem 4'. - The verification report's numerical table (Section 9 of this document) should be included as a verification table.

**Direct inclusion possible**: - Notation tables (Section 0.1 of this document). - Assumption catalog (Section 0.2). - Lemma statements (not full proofs, just statements). - Numerical verification table.

**Needs LaTeX-only reformatting**: - Theorem proofs (convert □ environments to align environments). - Long derivations (Lemma A Stirling expansion, Lemma D adaptive threshold derivation).

### 9.3 Cross-References

- Theorem 1 references Theorem 3 for necessity justification.
- Theorem 2 references Proposition 6 for practical diagnostic.
- Theorem 4' references Lemma B from Theorem 1's proof chain.
- Theorem 5 references Theorem 2 as the negative counterpart.
- All theorems reference the assumption catalog (A1-A6).

### 9.4 Estimated Total SI Pages

| Section | Pages |
|---|---|
| A.1 Notation and Assumptions | 2 |
| A.2 Theorem 1 | 6 |
| A.3 Theorem 2 | 5 |
| A.4 Theorem 3 | 4 |
| A.5 Theorem 4' (Lemmas A-F) | 12 |
| A.6 Theorem 5 | 8 |
| A.7 Proposition 6 | 3 |
| A.8 Numerical Verification | 1 |
| **Total** | **41** |

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## 10. References

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*End of THEOREMS_UNIFIED.md – Single source of truth for the SCX mathematical theory chain.*